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Copenhagen Versus the Universe

"If a poll were conducted among physicists," Bryce DeWitt wrote in 1970, "the majority would profess membership in the [Copenhagen] camp, just as most Americans would claim to believe in the Bill of Rights, whether they had ever read it or not." DeWitt had managed to convince the editor of *Physics Today,* the monthly magazine for members of the American Physical Society, to publish an article on the foundations of quantum physics. In a sign of the times, the editor, Hobart Ellis Jr., had not been particularly difficult to convince. "For a long time I personally have been dissatisfied with the apparent contradictions that physicists appear to be ready to live with in quantum mechanics and its interpretation," he wrote to DeWitt. "I think a general review of different interpretations of quantum mechanics without special emphasis on any one would be of interest."

DeWitt's article, "Quantum Mechanics and Reality," did review several interpretations. But DeWitt made his own opinions quite clear. "The Copenhagen view promotes the impression that the collapse of the [wave function], and even the [wave function] itself, is all in the mind," he wrote. "If this impression is correct, then what becomes of reality? How can one treat so cavalierly the objective world that obviously exists all around us?" When dealing with a system in a quantum superposition, like Schrödinger's cat, DeWitt said most physicists "conceive the [measurement device] to have entered a kind of schizophrenic state in which it is unable to decide what value it has found for the system," a living cat or a dead one. This problem, he concluded, was not resolved by the Copenhagen interpretation.